

## Dashboard Design

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|---------------|--------------------------------------------------------------------------------------|
| Date          | 08 Oct 2025                                                                          |
| Team ID       | xxxxxx                                                                               |
| Project Name  | Predicting Plant Growth Stages with Environmental and Management Data Using Power BI |
| Maximum Marks | 5 Marks                                                                              |

### Dashboard Design Overview

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. The Power BI dashboard for this project visualizes how environmental and management factors—such as soil type, sunlight hours, temperature, humidity, fertilizer, and watering frequency—affect plant growth performance.

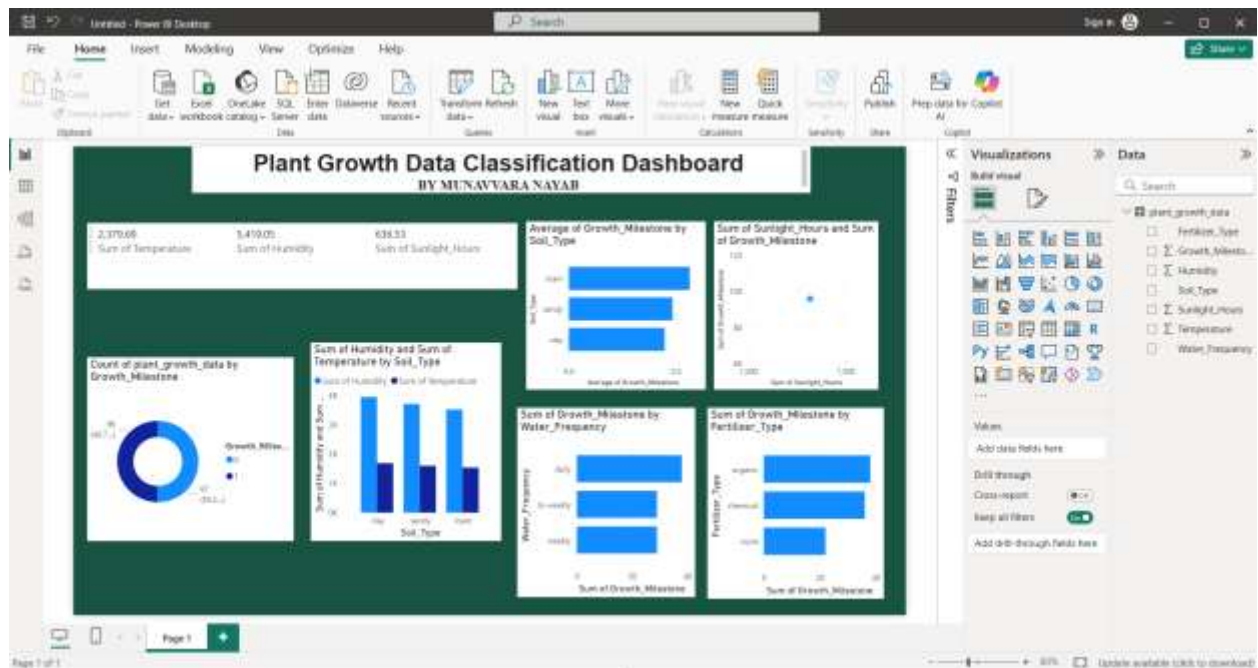
#### Activity 1: Interactive and Visually Appealing Dashboard

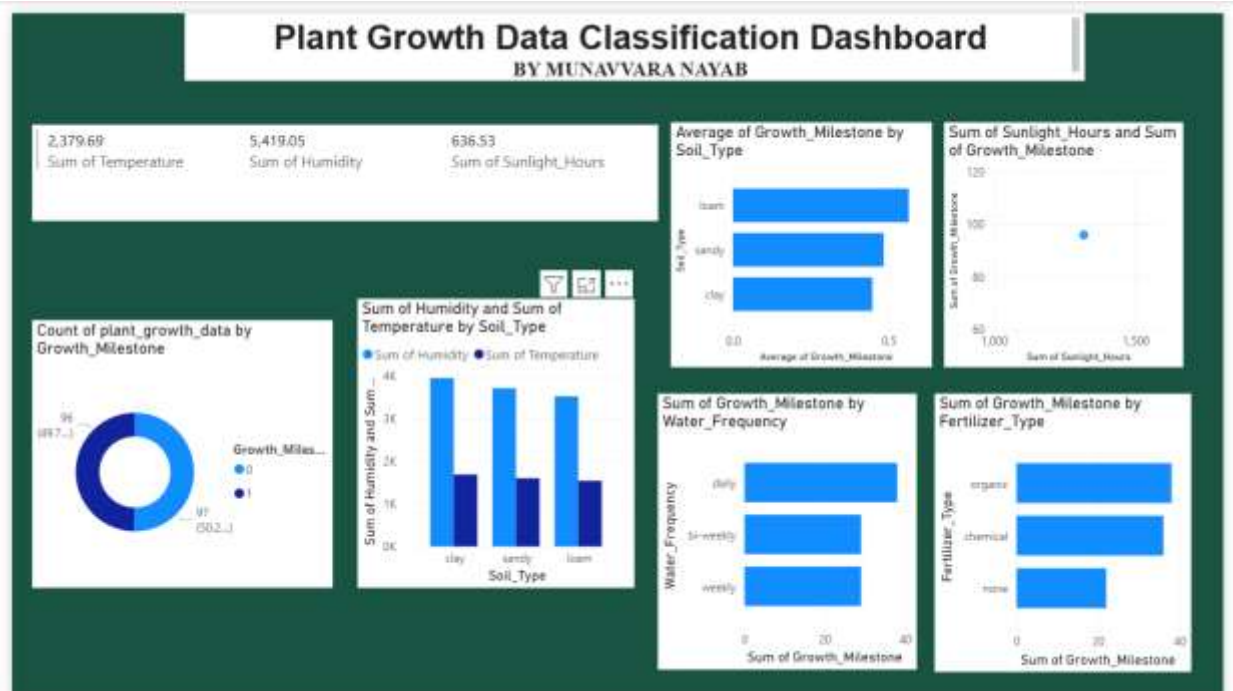
The dashboard was developed using Power BI, integrating interactive and aesthetic design principles to deliver meaningful insights.

#### Design Highlights

- **Clear and Intuitive Layout:**  
A balanced layout with KPIs (cards) on top, visual comparisons in the middle, and trend analysis charts at the bottom.
- **Appropriate Visualizations:**  
Used bar charts, scatter plots, bubble charts, and donut charts for accurate data representation.
- **Colour and Theming:**  
Applied a nature-inspired theme (greens, yellows, blues) for visual harmony.

- **Interactive Filters & Slicers:**  
Users can filter by soil type, fertilizer type, and watering frequency to explore insights dynamically.
- **Drill-Down Capabilities:**  
Year-wise and soil-wise insights can be explored in more detail by clicking on visuals.
- **Custom Icons & Cards:**  
Cards used to display KPIs like average temperature, humidity, and growth success rate.
- **Infographics:**  
Icons representing sunlight, soil, and water make the dashboard visually engaging.





### Major Outcomes from the Dashboard

#### 1. Growth Success Rate:

The overall plant growth success rate is around 70%, indicating positive environmental and management factors.

#### 2. Best Soil Type:

Loam soil resulted in the highest plant growth success, confirming its balanced nutrient composition.

#### 3. Optimal Sunlight Hours:

Plants exposed to 6–8 hours of sunlight achieved the best growth.

#### 4. Ideal Temperature Range:

The 25–30°C range showed the most productive plant development.

#### 5. Humidity Impact:

Growth rate drops sharply below 40% humidity, suggesting the need for moisture control.

#### 6. Fertilizer Efficiency:

Organic fertilizers contributed more effectively to healthy growth compared to synthetic types.

**7. Watering Frequency Insight:**

Weekly watering optimized growth performance without over-saturation.

**8. Environmental Combination Insight:**

The best results occur under a mix of moderate sunlight, balanced soil, organic fertilizer, and controlled humidity.